Hard thermodynamic constraints and the temperature extrapolation of learned water isotherms in metal-organic and covalent organic frameworks: a pre-registered test

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Abstract

Pending result (abstract). Written from the confirmatory verdicts (results/verdicts_confirmatory.json) after the analysis.

Introduction

Water sorbents built from metal-organic frameworks (MOFs) and covalent organic frameworks (COFs) are studied for atmospheric water harvesting[1, 2] and for adsorption heat pumps and chillers[3, 4]. These devices run between temperatures: a MOF-801 cooling cycle adsorbs at 30 °C and regenerates at 80 to 85 °C[5]. Its design needs the uptake at temperatures where no isotherm was measured, and the isosteric heat, which enters the cycle model[4]. Both rest on few temperatures. A practical guide asks for isotherms spaced by 20 K or less for a Clausius-Clapeyron heat and advises against extrapolation beyond the measured range[6]. For Mg-SU-102, isotherms were measured from 15 to 55 °C, but only those from 25 to 40 °C had enough points for the heat[7].

The standard temperature model for water in MOFs is the characteristic curve: the uptake is written as a function of the adsorption potential $A = RT \ln(p_{\mathrm{sat}}/p)$, the isotherms of all temperatures are expected to fall on one curve, and that curve is used to model cycles[8]. Within this approach the isosteric heat is the vaporisation enthalpy plus A[4]. Invariance is usually shown by overlaying curves, without an error number, as for MIL-160 from 20 to 50 °C[1] and Al-fumarate from 298 to 333 K[9]. The Mg-SU-102 curves shift slightly towards lower energies as the temperature rises[7], and MOF-303 loses a large part of its capacity between 25 and 45 °C[10]. Statistical thermodynamics gives a characteristic curve when added sorbate contributes purely energetically, and a temperature-dependent isotherm when sorbate number correlates with entropy[11]. The other common route, Langmuir or Sips sites with temperature-dependent affinities fitted to all temperatures at once[4, 12], has its own defects. Sips and Freundlich terms have no finite slope at zero pressure and their heats diverge at low coverage, dual-site Langmuir heats can show an unphysical maximum, and models that fit equally well disagree on the heat wherever they extrapolate[12].

Machine-learned isotherm models add flexibility but little thermodynamics. In two recent examples temperature is a plain input: one network predicted methane isotherms of 17,644 MOFs from 278 to 338 K and was extrapolated to 268 and 358 K[13]; another predicts water isotherms from pressure and structural properties[14]. Where heats appear, they are separate outputs. A neural ordinary differential equation learns $\mathrm{d}q/\mathrm{d}\ln p$ at one temperature, observes monotonicity without imposing it, and predicts the heat with a second network[15]. A zeolite study joined a learned loading at 300 K and an independently learned heat through an integrated Clausius-Clapeyron step with a temperature-independent heat[16]. When physics enters, it enters as a penalty. A network for hydrogen sorption adds monotonicity as a weighted loss term and reports

98.6 % monotonicity satisfaction, so 1.4 % of held-out cases violate it[17]. A penalty-trained network is not provably monotone; monotonicity can only be certified afterwards[18].

A different construction is possible. The grand (immersion) potential of the adsorbed phase, $\Omega = -RT \int n \, \mathrm{d} \ln p$, has the isotherm as its derivative with respect to $\ln p$[19, 20]. A model can learn this one scalar as a function of $\ln p$ and T. If it is convex in $\ln p$, the isotherm is monotone for every parameter value, and input-convex networks give convexity by construction[21]. A potential linear in p at low pressure gives a finite Henry slope, which Sips terms lack[12]. The isosteric heat then follows by Clausius-Clapeyron from second derivatives of the same function and is defined wherever $\mathrm{d}q/\mathrm{d} \ln p$ is positive. Neighbouring fields already learn potentials this way. A learned free energy gives fluid properties as its derivatives, with exact Maxwell relations[22]. In mixture thermodynamics, a graph network for the excess Gibbs energy replaced a Gibbs-Duhem penalty[23] and needs no extra loss terms[24], and hard-constraint networks build the Gibbs-Duhem relation into the architecture[25] and take the excess enthalpy from the temperature derivative of a learned excess Gibbs energy[26]. In adsorption, a learned density functional has produced an isotherm of a Lennard-Jones fluid in a model pore[27]. In the sources we searched, no model learns such a potential from measured isotherms of real sorbents or tests it on water.

Hard constraints make a model admissible, but not necessarily accurate. For phase-transition enthalpies, a hard thermodynamic constraint raised the error by more than 10 $\mathrm{kJ\,mol^{-1}}$, while a soft one kept the accuracy[28]. Two parts of the construction can also act separately: the shape constraint fixes monotonicity and the Henry limit, while the temperature structure (a van't Hoff affinity per term, a free dependence on $1/T$, or a characteristic curve) decides how the isotherm moves between temperatures. The open question is which of the two, if either, makes extrapolation in temperature better than penalties, generic networks and classical models.

The present study therefore asks whether hard constraints on a learned adsorbed-phase potential make its temperature extrapolation of measured water isotherms better than penalties, generic networks and classical models. It does so in a pre-registered factorial design that crosses the shape constraint (none, penalty or hard) with the temperature structure (free in $1/T$, van't Hoff per term, or a temperature-invariant characteristic curve). The learned potential is the cell with hard constraints and van't Hoff terms. It is compared with its penalty and unconstrained twins, the hard-constrained cells of the other two temperature structures, a generic neural network and a dual-site Sips model with van't Hoff affinities. The data are 13 real sets of water isotherms measured at three or more temperatures, 5 on MOFs and 8 on COFs. Each model predicts a material's highest and lowest temperatures from the others. The design, tests and words of every verdict were fixed before any confirmatory fit, including words for a classical model that wins and for models equivalent within 25 %. Every defect found in the design, code or data is recorded in a public correction ledger. None of the 13 materials has a calorimetric water heat; our search found one verified value for a MOF with multi-temperature isotherms, a single integral heat for MOF-801[29], and none for any COF. We therefore test the heat on synthetic truths with exact heats, built on each material's own temperatures and pressures.

Pending result (contributions). Numbered contributions, each with its headline number, evidence label and section pointer (lessons file, addendum on the pinn-machining style guide), written from the confirmatory verdicts.

Results

Pending result (results-design). Figure 1 (paper/figures/fig_design.pdf): the factorial, the tasks and the evidence label of every comparison. Fits completed and failed per cell, from results/confirmatory_runs.shard*.jsonl.

Pending result (results-H1). H1b (primary, non-inferiority) to H1e: R with exact 95 % and 90 % intervals, Holm-adjusted p and verdict words, from results/verdicts_confirmatory.json (paper/table_verdicts.py); Figure 2a forest (paper/fig_verdicts.py). H1a descriptive.

Pending result (results-H3). H3 admissibility: violating fits per cell over the E-high and E-low fits, Wilson intervals, the pre-registered prediction (V-pen any class but none and P none); Figure 2b. The P bug check (P_bug_fits).

Pending result (results-H2). H2 isosteric heat on the synthetic truths (results/synthetic_h2 confirmatory runs), the 15 tests of PREREG 7.1.

Pending result (results-descriptive). Descriptive analyses fixed in PREREG 6.4: I-mid, seed spread, the MLP, F-rel, P-K, subgroups; the sensitivity variants (scripts/sensitivity.py verdicts NAME, seven variants) and MOF-801.

Discussion

Pending result (discussion). What the verdicts mean for hard constraints against penalties (accuracy against admissibility), the temperature structure, limitations (n = 13, digitised data, no calorimetric heats, two training temperatures in E tasks for most materials) and what is not claimed (PREREG section 9).

Methods

Pre-registration

The design, the data, the models, the tuning rules, the metrics, the statistical tests and the words of every verdict were fixed in a pre-registration and frozen before any confirmatory fit. The freeze is a public, tagged commit (`prereg-p3a-freeze`) whose manifest holds the SHA-256 of 71 files (the text, the data, the tuned configurations and the fits behind them, the synthetic-world and power results, and every script that builds, fits or scores, including an independently written second scorer) and the package versions; the confirmatory scripts refuse to run if any of them differs. Every defect found in the design, code or data, before or after the freeze, is listed with its cause and fix in a public ledger.

Data

Water adsorption isotherms of MOFs and COFs measured at several temperatures on one sample were collected from the primary literature and from the NIST adsorption database. A material entered the confirmatory set if, at three or more temperatures, its adsorption isotherm had at least 8 points with $0 < p/p_{\mathrm{sat}} \leq 0.90$, its figure passed the digitisation quality gates, its temperatures were unambiguous, and its curve was covered: no gap above 0.25 in p/p_{sat}, and a first point at $p/p_{\mathrm{sat}} \leq 0.10$ or at an uptake of at most 0.10 q_{ref}. Where a paper gave both measured isotherms and curves derived from them, the measured isotherms were used. The confirmatory set has 13 materials, 5 MOFs and 8 COFs; every decision and its reason is in the inclusion log. Four further materials form a development set used only to choose hyperparameters.

Isotherms published only as figures were digitised before any model was fitted. Vector figures were read from the PDF drawing commands, with marker centres mapped through the axis ticks; raster figures by template matching with the figure's own marker footprint, with tick marks located to sub-pixel precision and an open-marker veto against desorption points. No physical prior was used to filter points. The method was validated against published numeric data. On Mg-SU-102 the vector extraction recovered 198 of 198 points with a root-mean-square uptake error of 7×10^{-6} g g^{-1}. On raster figures of HKUST-1 the root-mean-square uptake error was 0.11 to 0.19 % of the axis span, or 0.0012 to 0.0034 of the isotherm's maximum, with no false detections.

Only adsorption branches with $p/p_{\mathrm{sat}} \leq 0.90$ were used. Relative pressures were converted with the IAPWS saturation pressure, absolute pressures were used as reported, and uptakes

were converted to mol kg^{-1}. The scale q_{ref} of a material, its largest uptake in this range over all temperatures, is used only for scoring.

Models

Every model is a function $q(s, T)$ of $s = \ln(p/\mathrm{Pa})$ and temperature, and the isosteric heat of every model is computed by the same code, $Q_{\mathrm{st}} = -RT^2(\partial q/\partial T)/(\partial q/\partial s)$, by automatic differentiation.

The potential model (P) learns the reduced spreading pressure

$$\psi(s,T) = \sum_k \frac{c_k(T)}{g_k}\,\mathrm{softplus}\big(g_k\,(s-\theta_k(T))\big), \qquad q^* = \frac{\partial \psi}{\partial s} = \sum_k c_k(T)\,\sigma\big(g_k\,(s-\theta_k(T))\big), \quad (1)$$

with capacities $c_k > 0$, Henry terms ($g_k = 1$) and cooperative terms ($g_k \geq 1.5$), van't Hoff positions $\theta_k(T) = \theta_{k0} - (Q_k/R)(1/T - 1/T_{\mathrm{ref}})$ and an adsorbate-expansion factor. For every parameter value, q^* increases strictly with p, q^*/p tends to a positive Henry constant as $p \to 0$, and the heat is defined everywhere.

The models form a factorial of shape constraint (none, penalty, hard) and temperature structure: a free network in $1/T$ (F), van't Hoff per term (V), or the temperature-invariant characteristic curve in A (A). In the V and A columns, the unconstrained and penalised models have the function class of P or of the hard characteristic curve without the constraints, the same initial function and the same tuned structure; the penalty models add monotonicity and (except for the characteristic curve) a scale-free Henry-limit penalty on collocation points that reach the held-out temperature. A dual-site Sips model with van't Hoff affinities (C1) is the classical comparator. A multilayer perceptron on $(\ln p/p_{\mathrm{sat}}, 1/T)$ and a Kirchhoff variant of P with a heat capacity per term are reported descriptively.

Fitting and tuning

Each model was fitted by Adam followed by L-BFGS on the mean squared error of q scaled by the largest training uptake, in double precision, from 5 seeds; the fit with the lowest training loss was kept, never selected on held-out error. Every scale of a fit came from the training temperatures only. Hyperparameters were chosen on the development set alone, in two stages: the models that own a structure tried up to 12 configurations each, and the unconstrained and penalised models inherited the chosen structure, the penalty models tuning only the penalty weight over 10^{-4} to 100.

Tasks and outcomes

For a material with temperatures $T_1 < \cdots < T_k$, the task E-high trains on $T_1, \ldots, T_{k-1}$ and predicts T_k; E-low trains on $T_2, \ldots, T_k$ and predicts T_1. The primary outcome is the held-out error $E = \mathrm{RMSE}(q_{\mathrm{pred}} - q_{\mathrm{obs}})/q_{\mathrm{ref}}$, summarised per material as the mean of $\log_{10} E$ over the two tasks. Physical violations were counted on a dense grid: a fit violated if an isotherm fell by more than 2 % of q_{ref} below its running maximum, or went below -2 % of q_{ref}, inside the domain where the penalties act; violations were counted over the fits of the two extrapolation tasks. Because no material here has a calorimetric water heat, the heat was tested on synthetic isotherms with exact heats, built on every material's own temperatures and pressures from five truth families, with noise of 0.4 % of q_{ref}.

Isosteric heat on synthetic truths

The five truth families are a Langmuir site plus a cooperative Sips site with a heat capacity, a Dubinin-Astakhov curve, a Fowler-Guggenheim isotherm with lateral attraction, strong hydrophilic sites with a heat that falls steeply with loading, and a Sips isotherm whose capacity

falls with temperature (Supplementary Note 2). Each family is sampled on every confirmatory material's own temperatures and pressure grid, with shape parameters fitted to that material's middle-temperature isotherm and the thermodynamic parameters fixed in advance. P, V-pen, A-hard and F-none were fitted to the E-high and E-low tasks of every truth. The primary metric compares the heats at equal loading: at each held-out state where the model's own loading lies between 0.1 and 0.9 q_{ref}, the model's Q_{st} is compared with the truth's Q_{st} at the same loading and temperature, found by inverting the monotone truth, and the root-mean-square difference is taken in $\mathrm{kJ\,mol^{-1}}$. A task fails, and is ranked worst, if any compared heat is undefined or not finite, if fewer than 3 states fall in the window, or if the truth cannot reach the model's loading. The fifteen comparisons (three models against P in five families) are Holm-corrected as one family. The single calorimetric value found for a material with multi-temperature isotherms, MOF-801[29], is used only in a descriptive check; no claim about the heat of a real material is made.

Admissibility

For each model the violating fits are counted over the 26 fits of the two extrapolation tasks, with a Wilson 95 % interval; the counts over all tasks, adding the interpolation task, are reported beside them. The words follow a table fixed in advance: at least 20 % of fits when the lower bound reaches 20 %, no violations detected when no fit violates, fewer than 20 % when the upper bound is below it, and an uncertain rate otherwise. The pre-registered prediction is that penalties leave violations that the hard constraints remove: V-pen has at least one violating fit and P has none. It concerns V-pen at the penalty weight chosen for accuracy on the development set ($\lambda = 0.001$), not penalties at every weight. By construction P cannot violate; a violating fit of P would be a bug and halts the analysis.

Descriptive and sensitivity analyses

Fixed in advance and reported without a hypothesis: the interpolation task; P against V-none (the pure effect of the constraints at equal structure), against F-hard (rigid against free temperature dependence), against the multilayer perceptron and against the Kirchhoff variant of P; MOFs and COFs separately; the error against the distance of the held-out temperature from the nearest training temperature; and the spread of the characteristic curve across temperatures. The H1 comparisons were repeated on the datasets given by earlier versions of the coverage rule and by a 7-point rule (6, 9, 23 and 14 materials), on Co_2Cl_2(BTDD) from its characteristic curves, and on two perturbed copies of the data in which every digitised isotherm is shifted in p/p_{sat} by ± 0.002, the largest systematic shift measured in the digitisation benchmark.

Software

All fits ran in double precision on a CPU with Python 3.12.10, PyTorch 2.12.0, NumPy 2.4.6, SciPy 1.17.1 and pandas 3.0.3; the versions are part of the frozen manifest.

Statistics

The unit is the material ($n = 13$). For each comparison of P with a model X, $d_m = e_m(\mathrm{P}) - e_m(\mathrm{X})$ was tested with the exact two-sided Wilcoxon signed-rank test, with Holm correction over the four confirmatory comparisons (penalty model, characteristic curve, generic network, dual-site Sips). The effect is the Hodges-Lehmann estimate with exact distribution-free 95 % and 90 % intervals, reported as the error ratio $R = 10^d$. A failed fit was ranked worst, never dropped. The words of each verdict follow a table fixed in advance, with a smallest effect of interest of 25 % ($R = 0.8$ or 1.25). The one declared primary claim, that the hard constraints cost no more than 25 % in accuracy against the penalty model, was judged by non-inferiority: the upper

exact 95 % bound of R below 1.25 (one-sided 2.5 %), with the 98.75 % bound reported beside it. At 13 materials its exact size at the margin is 0.024 (0.005 for the 98.75 % bound). Before the freeze, the power of each verdict was simulated with the analysis code itself, and the whole pipeline was run on five synthetic worlds, with an expected outcome written for each before it ran, to show which verdicts can fail (Supplementary Note 1). Three of the expectations were missed and are reported as missed, and the comparison of a free with a van't Hoff temperature dependence was made descriptive because it could not fail.

Data Availability

The pre-registration, the freeze manifest, the corrections ledger, the digitisation provenance and all results are public at `https://github.com/abdulhamidbatayhi123/ai-isotherm-thermo`; an archived release with a DOI will be cited. Digitised point lists from sources whose licence does not permit redistribution are kept locally; their provenance, fitted parameters and predictions are published.

Code Availability

All code is in the same repository and archive.

Use of AI tools

Pending result (ai-statement). The author's saved wording (AI as a tool; the author designed the study, made every decision and takes responsibility), adapted to the target journal's policy.

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